



The benefits of cloud computing

Compared to traditional on-premises infrastructure, cloud computing offers a fundamental shift in how resources are accessed and delivered. This shift results in powerful advantages across speed, scale, and security, providing organizations with the ability to innovate faster and pivot resources instantly. Ultimately, the cloud allows businesses to move beyond managing physical hardware and focus instead on maximizing business value.

To learn more, expand each of the ten items.

It's **scalable**.



Cloud computing gives organizations access to scalable resources and the latest technologies on-demand, so they don't need to worry about capital expenditures or limited fixed infrastructure. This can significantly accelerate infrastructure deployment time.

It's **flexible**.



Organizations and their users can access cloud services from anywhere with an internet connection; scaling services up or down as needed to meet business requirements.

It's **agile**.



Organizations can develop new applications and rapidly get them into production, without worrying about the underlying infrastructure.

It's **fast**.



Google Cloud is built upon Google's own global private fiber network, which is optimized to travel through the shortest, fastest, private routes possible before it touches the public internet.

It has **global reach**.



Cloud providers operate vast networks of data centers all over the world, organized into regions and availability zones. This infrastructure allows organizations to deploy applications and store data closer to their end-users in different countries.

It offers **high availability**.



To reduce downtime, Google Cloud replicates infrastructure across multiple, isolated availability zones within a region, ensuring 99.999% availability.

It offers **strategic value**.



Because cloud providers are continuously innovating and offering their breakthroughs as services to customers, organizations can get more competitive advantages—and a higher return on investment—than if they'd invested in soon-to-be obsolete technologies. This gives organizations a strategic advantage by working with the latest technology advancements.

It powers **data-driven insights**.



Cloud computing provides a serverless, instantly scalable data platform that integrates advanced machine learning directly with massive datasets, enabling organizations to move beyond simple reporting to real-time, predictive decision-making.

It's **secure**.



Cloud computing security is recognized as stronger than that in enterprise data centers, because of the depth and breadth of the security mechanisms and dedicated teams that cloud providers implement.

It's **cost-effective**.



No matter which cloud computing service model organizations implement, most cloud providers offer flexible consumption models so you only pay for what you use. They don't need to overbuild data

center capacity to handle sudden spikes in demand or business growth, and they can deploy IT staff to work on more strategic initiatives.

Customer examples

Loblaw, HSBC, and the American Cancer Society

What does Canada's largest food retailer have in common with a global bank and a leading cancer-fighting organization? They're all harnessing the power of the cloud to innovate.

Let's explore how **Loblaw**, **HSBC**, and the **American Cancer Society** are making an impact.

Click play to start the video.



Loblaw, HSBC, and the American Cancer society

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Transcript

Cloud adoption has made a positive impact on some of the world's leading companies across various industries.

Let's start with an example of how the cloud provided flexibility and improved performance for the Canadian food and pharmacy leader Loblaw. Loblaw is Canada's largest retailer, with nearly 2,500 corporate, franchised, and associate-owned locations and nearly 200,000 full- and part-time employees. The Vice President of Technology at Loblaw explained: "We want our tech talent focused on creating better experiences for our customers, not maintaining infrastructure."

Loblaw took a lift-and-shift approach to accelerate the initial migration to Google Cloud. This means they focused on moving their existing virtual machines on-premises to Compute Engine instances in the cloud without needing to redesign them. However, they designed the cloud architecture to be easily converted to using Google Kubernetes Engine for automated deployment and scaling later. This shows how an open cloud can grow with an organization's needs. With a more responsive ecommerce site and the ability to handle more traffic without affecting the customer experience, Loblaw could run marketing promotions and generate additional revenue. With these improvements, they expect to reclaim 50% of their Site Reliability Engineers' time to focus on innovation.

Now let's shift our focus to another example, this time on how the cloud provides scalability and cost reduction. HSBC is one of the world's largest banking and financial services organizations, serving more than 40 million global customers from offices in 64 countries and territories. Committed to a cloud-first strategy, in January 2021, HSBC embarked on an ambitious project to enhance and future-proof their risk management on the cloud. Their previous on-premises system was not capable of meeting future regulatory and business demands. HSBC built a cloud-native risk management solution that boosts calculation speeds to be ten times faster while lowering costs. This equated to three billion calculations per second. The power of a data cloud is that it almost has unlimited resources to process large volumes of data and reduce time to insights.

A Global Head of Traded Risk Technology at HSBC explains, "We knew that a cloud-native solution gave us the ability to scale and run at a reduced cost. We did a proof of concept using Google Cloud, and we quickly realized that this could be very successful." HSBC built a cost-effective platform that is faster and more efficient while meeting their regulatory and compliance requirements.

And in a final example, let's look at how the cloud has brought agility and valuable insights, while maintaining trust, to an organization. The American Cancer Society is a community-based volunteer health organization dedicated to eliminating cancer as a major health problem. Their mission is to free the world from cancer by funding and conducting research, sharing expert information, supporting patients, and spreading the word about prevention.

Among women, breast cancer is the most commonly diagnosed type of cancer. Yet, if detected early, it's also one of the most survivable cancers. Mia M. Gaudet, PhD, is the Scientific Director of Epidemiology Research at the American Cancer Society. Through her research, she's obtained over 1,700 high-resolution tissue images from participants diagnosed with breast cancer. This valuable data could help them discover factors that could lead to a cancer diagnosis and improve survival rates. The challenge was to identify novel patterns in digital images of breast cancer tissues to potentially improve patient outcomes.

Their research group partnered with Slalom, a Google Cloud Premier Partner, and sought to combine machine learning-powered insights with cloud computing performance to improve timeliness and accuracy. By building a machine learning pipeline using Google Cloud AI Platform, now called Gemini Enterprise Agent Platform, they trained models for AI image analysis of tissue scans to find cancer indicators. The team achieved 12 times faster image analysis with enhanced quality and accuracy by removing human limitations. Dr. Gaudet said, "The ability to perform image analysis by using deep learning for

epidemiologic breast cancer studies opens a new frontier of research, and Google Cloud makes it easier. We're excited about what we'll find."

The American Cancer Society is now equipped with processes and a cloud infrastructure that will be reusable on similar projects, providing a foundation for future work.

Reflection

A regional bank is weighing the high cost of maintaining a legacy on-premise system against the effort required to migrate to the cloud to automate their fraud detection. They're concerned about compatibility with their current proprietary data warehouse and future flexibility.

Your task: Consider the concept of open cloud. How does adopting an open infrastructure help this bank minimize the risks of vendor lock-in and interoperability issues? Why is this flexibility particularly critical for a bank managing strict compliance records?

Reflect on the scenario above and type your thoughts in the box below.

In a public cloud scenario the bank could use the

Good thoughts! An open cloud approach empowers the bank to integrate existing systems with innovative new tools while maintaining complete provider flexibility. This strategy is vital for navigating strict banking regulations, ensuring data remains portable and easily auditable for compliance. By leveraging shared standards, the bank avoids vendor lock-in and keeps its fraud detection systems agile, secure, and ready for future technological shifts.

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